

Anamol Shrestha

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PROFESSIONAL SUMMARY

Principal Engineer specializing in high-throughput event stream processing and distributed systems architecture with 8+ years of production experience. Designed Kafka producer/consumer topologies guaranteeing exactly-once delivery at 100K+ events/day with sub-50ms latency for \$2B+ daily trade settlements - with message deduplication, transactional producers, and custom retry mechanisms hitting 99.95% message delivery reliability. Deep expertise in streaming latency optimization, fault-tolerant microservices, and real-time pipeline design across regulated financial and enterprise environments. Seeking a Principal IC role with high ownership to solve hard streaming infrastructure problems at scale.

WORK EXPERIENCE

Fidelity Investments

Aug 2024 - Present

Principal Full Stack Engineer - OpsWorX Platform

Westlake, TX

- Engineered the **SDK event management framework** for real-time cross-application event stream processing - reducing data propagation latency by **40%** and inter-app communication failures by **60%** across 50+ hosted applications and thousands of daily active users.
- Architected and led development of Java Spring Boot backend APIs powering the OpsWorX interoperability platform, serving 20+ internal and partner engineering teams with **99.9% uptime** in production.
- Drove architectural decision-making for migration of 15+ legacy WPF/Prism applications to a modern **microservices-based** hosting platform - delivering zero-downtime cutover and reducing deployment cycle time by 35%.
- Designed and enhanced **cross-application event streaming APIs** via io.Connect, enabling real-time data sharing, window management, and global notifications with **sub-50ms event propagation**.
- Standardized REST API integration patterns across 5+ business units, reducing partner team onboarding time by **50%** and eliminating platform fragmentation.
- Mentored 5+ engineers through design reviews and hands-on guidance; established coding standards adopted platform-wide, accelerating feature delivery velocity by 25%.

SEI Investments

Mar 2022 - Aug 2024

Sr. Java Developer - Special Assets Platform

Remote

- Architected a fully **event-driven streaming architecture using Kafka and JMS** to orchestrate the end-to-end TBA Mortgage-Backed Securities trade lifecycle - processing **100,000+ trade events daily**, replacing synchronous flows and cutting settlement latency by **45%**.
- Designed **Kafka producer/consumer topology** with message-key-based **deduplication at scale**, transactional producers, and custom retry mechanisms - guaranteeing **exactly-once delivery** across Kafka and JMS at **99.95% message delivery reliability**.
- Developed Spring Boot REST APIs integrated with **high-throughput event streams** for real-time trade status visibility, supporting 8+ consuming services with **sub-100ms response times** under peak load.
- Key contributor to financial instruments platform supporting **\$2B+ in daily trade volume** across complex asset classes - options, capital calls, MBS, contributions, distributions.
- Onboarded a net-new TBA MBS asset class end-to-end - from trade creation through settlement - expanding platform capabilities with a **fault-tolerant, low-latency pipeline** built from first principles.

American Airlines

Aug 2018 - Mar 2022

Software Engineer - Inbound IVR Platform

Fort Worth, TX

- Maintained and modernized core Java APIs supporting IVR systems handling **5M+ customer interactions monthly** at 99.9% availability across American Airlines' global customer base.
- Delivered the IVR Containment Deeplink Project - introducing SMS-based self-service, achieving **12% customer containment** and deflecting **600,000+ calls annually**, saving an estimated \$3M+ in operational costs.
- Enhanced the Proactive Greeting Service using **multithreading and real-time event processing**, delivering personalized IVR experiences processing **1M+ events monthly** with sub-200ms response times.
- Led platform transition from Java monolith to **microservices on PCF/AWS** - decomposing 12 tightly coupled services, improving deployment frequency by 4x and reducing production incidents by 30%.

EDUCATION

University of Cumberlands MS, Information Technology	December 2022
University of Louisiana at Monroe BS, Computer Science BS, Mathematics	April 2018

TECHNICAL SKILLS

- Messaging & Streaming:** Kafka, Kafka Streams, JMS, RabbitMQ, ActiveMQ - high-throughput event stream processing, exactly-once delivery, transactional producers, message deduplication at scale
- Distributed Systems:** Microservices, Event-Driven Architecture,DDD, CQRS, Saga Pattern, High-Concurrency, Low-Latency Pipeline Design, Fault-Tolerant Systems
- Languages:** Java, TypeScript, Python, JavaScript, C#, SQL, Shell Script, Groovy
- Backend & Frameworks:** Spring Boot, Spring Cloud, Spring Batch, Node.js, Express.js, gRPC, REST, GraphQL
- Cloud & Infrastructure:** AWS (EC2, S3, Lambda, SQS, SNS, RDS), Azure, GCP, Docker, Kubernetes, Helm, Terraform, Istio, PCF
- Databases:** PostgreSQL, Oracle, MySQL, MongoDB, Redis, Elasticsearch, DynamoDB
- Observability & CI/CD:** Datadog, Dynatrace, Splunk, Prometheus, Grafana, ELK Stack, Jenkins, GitHub Actions, ArgoCD, SonarQube